# Lightweight Restraints for Powerful Agents: Effective AI Control Monitors at a Fraction of the Cost

**Jeremias Ferrao**[*]
Independent

**Abhinav Rajeev Kumar**
Independent

## Abstract

[TODO: Abstract: problem (monitoring on the critical path of agent actions), method (binary-logit distillation from a frontier teacher into small open-weight monitors), headline results, and what is released.]

## 1 Introduction

[TODO: Motivate cheap, fast monitors: the safety tax of frontier monitors on the critical path of agent actions.]

[TODO: State contributions as a short list: training recipe, evaluation, analysis, released models and serving stack.]

## 2 Related work

[TODO: Trained control monitors (prior work before ours), rationale distillation, white-box probes, monitor-directed attacks [Terekhov et al., 2025].]

## 3 Training and serving Gleipnir monitors

[TODO: Task formulation: trajectory in, single decision token out.]

### 3.1 Binary-logit distillation

[TODO: Teacher probability on the decision token as a soft target; contrast with severity scores and rationale SFT.]

### 3.2 Data and training setup

[TODO: Data sources and grouped holdouts, LoRA configuration, model sizes, checkpoint averaging.]

### 3.3 Serving and cost

[TODO: Main serving decisions: single prefill pass, low-precision weights with QAD for the 4B, last-token final-layer attention, host-path changes. Cost per 1,000 evaluations at on-demand and committed-use B200 rates. Score change against BF16 serving. Released artifacts. Throughput per change goes in the appendix.]

[*]Corresponding author: j.lino.ferrao@gmail.com.

## 4 Main evaluation

[TODO: Metric definition (pAUROC@20 and supporting metrics), in-distribution and OOD suites, baselines.]

### 4.1 Cost–performance frontier

[TODO: Frontier figure and table with bootstrap intervals.]

## 5 Held-out and adversarial evaluation

[TODO: Held-out suites and SleightBench (stripped view); monitor-directed injections.]

## 6 Probing at half depth

[TODO: Hard-label then soft-target probes on a fixed probe family, compared with fine-tuning on quality and cost. Frame as representation analysis, not a deployment alternative.]

## 7 Limitations

[TODO: Scope of evaluation, teacher dependence, known failure modes and adversarial limits.]

## 8 Conclusion

[TODO: Summary and future work.]

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## References

Mikhail Terekhov, Alexander Panfilov, Daniil Dzenhaliou, Caglar Gulcehre, Maksym Andriushchenko, Ameya Prabhu, and Jonas Geiping. Adaptive attacks on trusted monitors subvert AI control protocols. *arXiv preprint arXiv:2510.09462*, 2025.

## A Additional results

[TODO: Per-source results, hyperparameters, prompts and reproducibility details.]